



Dr Avishek Adhikary

Asst Prof, IIT Bhilai
PhD, IIT Kharagpur
Senior Member, IEEE



8 June 1985



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Research Profile Highlights

- 6 Projects (funded by MeitY, SERB, IBITF-DST) - total 293.64 Lacs
- **1 Patent Granted**, 2 Applied
- **18 Journal publications: 8 Q1, 9 Q2**
- 21 Conference publications
- 3 Book chapters
- Google Scholar *h* index **13**
- Google Scholar i-10 index 15
- **Total Citation 713**
- Highest Citation in a paper: 141
- Average Citation per paper: 20
- Average Impact Factor: 3.5

Teaching Profile Highlights

- 6.6 years as Faculty Member
- **Faculty member of 2 depts: EE & MT**
- Total 35 courses; 25 Theory, 10 Labs
- Total taught: 82 credits
- 2221 student enrollment.
- Maximum Class Strength: 176
- **Designed 16 Courses**
- **Average Course Feedback 4.38**
- **5 PhD**— 1 Graduated, 4 ongoing
- **11 MTech**— 4 Completed, 7 ongoing
- **5 BTech Hons**— 4 completed, 1 ongoing

Present Designations

Assistant Professor, Electrical Engg Department, IIT Bhilai, Since 17 Sep 2018;
Current Pay Level: 13A1 (Since 17 Sep 2024)

- Associate **Head of the Department, EE, IIT Bhilai**
- Invited **Member, Senate, IIT Bhilai**
- **Joint Faculty Member, Mechatronics Department, IIT Bhilai.**
- **Member, Institute Faculty Advisory Committee (IFAC), IIT Bhilai**
- **Member, Board of Studies, SSTU, Durg, Chhattisgarh**

Thrust Areas of Research

Research Lab: Applied Sensing and Control for Intelligent India (ASCI)

- Fabrication of Biosensors for in-home Devices and Smart Consumerism
- Sensor Data Analytics and Applied Machine Learning
- Sensor Electronics, Fractional Order Circuit Design
- Precision Agriculture, Fruit Quality Analysis
- Sensor Data Analytics for μ Single Point Incremental Forming and Under Water Laser Deposition
- Biomolecule Classifications: Hexose, Amine, Dopamine.
- Sensors for UAV and Mobile Robots

Education

2012-18	Ph.D. in Electrical Engineering Thesis Title: Four-Quadrant Fractors and Their Applications in Fractional Order Circuit Supervisor: Prof. Karabi Biswas and Prof Siddhartha Sen. Research Contribution: Design of a novel device, circuit synthesis, characterization and application; Signature innovation in the area of fractional order circuit theory. Date of Defence: 9 July 2018.	IIT Kharagpur
2010-12	M.Tech. in Electrical Engineering Specialization: Instrumentation Thesis Title: FPGA Based Digital Controller for Second and Higher Order Systems Supervisor: Prof. Karabi Biswas. CGPA: 9.02 Date of Defence: 8 May 2012	IIT Kharagpur
2003-07	B.E. in Electrical Engineering CGPA: 8.19	Jadavpur University
2001-03	Class XII Physice, Chemistry, MATHematics, Statsistics, Bengali and English Marks 89.5% (Among Top 80 in WBCHSE, 2003)	Hare School, Kolkata
1991-01	Class X Science, Mathematics, Humanities, Bengali and English Marks 89.7% (Among Top 50 in WBSEB, 2001)	Hare School, Kolkata

Previous Professional Experiences

2007-10	Assistant System Engineer in TCS Pvt. Ltd, 22/10/07 - 19/07/10 Database Manager, Programmer, Business Analyst, and Module Lead.	Role:
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Accolades and Achievements

2010	All India Gate Rank 22 in EE (Gate Score 934)
2012	Ranked 3rd in all MTech Specializations, EE Dept., IIT KGP.
2016	Best Research Paper Award in IEEE TechSym 2016
2017	Best Researcher Awards, RS Day 2017, IIT Kharagpur
2018	Award of Appreciation, in IEEE IM Student Contest 18, Huston, USA.
2023	Keynote speaker in 'Nigmatullin Readings 2023', Kazan, Russia.

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List of Sponsored Projects

Project 1

Title: Capacity building for Human Resource Development in Unmanned Aircraft System (Drone and related Technology)
Funding Agency: **MeitY - (SwaYan)**
Project Grant: **213.16 Lacs**
Role: **Chief Investigator**
Status: Ongoing
Duration: 5 Years
Start Date: 11 July 2022

Project 2

Title: Development of a Data-driven Fruit Ripening Detector based on Bio-Impedance of Senescent Fruits: Mangoes and Papayas
Funding Agency: **SERB - (CRG)**
Project Grant: **36.50 Lacs**
Role: **Principal Investigator**
Status: Ongoing
Duration: 3 Years
Start Date: 7 Feb 2024

Project 3

Title: Multipurpose Smart Agro-bot for Developing a Robust Data Model for Crop Condition Monitoring and Fertilizer Dispensing in Vertical Farming
Funding Agency: **IBITF - (EIR-PRAYAS)**
Project Grant: **27.30 Lacs**
Role: **Co-Principal Investigator**
Status: Ongoing
Duration: 1 Years
Start Date: 22 Mar 2024

Project 4

Title: Improving Financial Condition of Farmers through Real Time Precision Agriculture: Plant Disease Detection
Funding Agency: **IBITF, DST - (NM-ICPS)**
Project Grant: **8.20 Lacs**
Role: **Co-Principal Investigator**
Status: Ongoing
Duration: 1 Years
Start Date: 29 Mar 2024

Project 5

Title: Design and Fabrication of Fractor- The New Circuit Element
Funding Agency: **IIT Bhilai - (RIG)**
Project Grant: **5.53 Lacs**
Role: **Principal Investigator**
Status: Completed
Duration: 3 Years
Start Date: 2 Aug 2021

Patents

1. **A Adhikary**, S Bannerjee, S Sen, and K Biswas, "Two Electrode based System for Measuring Conductivity of Liquid Medium", Patent No 535231, Date of Filing 22/11/2017. Current Status: Patent **Granted**.
2. **A Adhikary**, A Bose, and A Tapadar, "A Paper Based Sensor for Urea Detection in Milk and a Method of Fabrication Thereof", Patent Application No.: 202521017877, Date of Filing 28/02/2025 Current Status: **Published**.
3. K Bandyopadhyay, **A Adhikary**, and A Khan, "A Portable Device for Underwater Dry Laser Welding", Patent Application No.: 202521012050, Date of Filing 12/02/2025 Current Status: **Published**.

Journal Publications (Post PhD)

1. A Tapadar, D Roy, and **A Adhikary**, "Ripe Stage Detection and Optimal Ripe Hours Prediction of a Banana using Modified Fractional Order Colpitts Oscillator", *IEEE Tran. Instrum. Measure.* **2025**, doi: 10.1109/TIM.2025.3565248. (Q1, IF: 5.6)
2. M R Samantaray, A Tapadar, S Das, N Chander and **A Adhikary**, "Low Temperature One-pot Synthesis of rGO Nanorod-PVDF Composite and Fabrication of a Thin Film Solid-state Fractional Order Device", *IEEE Tran. Nanotechnology*, **2024**. doi: 10.1109/TNANO.2024.3469973. (Q2, IF: 2.9)
3. A Bose, A Tapadar, S Khairnar, Y Mandawat and **A Adhikary**, "Modified Colpitts Oscillator with Lossy Fractor for the Biosensor with Single Cole Model and Its Application in Aqueous Urea Sensing", *IEEE Tran. Instrum. Measure.*, Vol 73, pp.1-8, **2024**. doi: 10.1109/TIM.2024.3458058. (Q1, IF: 5.6)
4. D Roy, A Ali, S Pal, and **A Adhikary**, "Classification of Sugar Samples from PCA-reduced Impedimetric Data using QD Model", *IEEE Trans. Instrum. Measure.*, Vol 73, pp.1-8, **2024**. doi: 10.1109/TIM.2024.3394472. (Q1, IF: 5.6)
5. A Tapadar, D. Roy, and **A Adhikary**, "Ripe Stage Classification of *Chausa* and *Banganapalle* Mango using Fractional Order Colpitts Oscillator", *IEEE Sensors Letter*, **2024**. doi: 10.1109/LSENS.2024.3433017. (Q2, IF: 2.2)
6. A Bose, A Tapadar, M. Kumar, and **A Adhikary**, "Detection of Urea in Adulterated Milk with 3% Fat using Disposable Paper Sensor", *IEEE Sensors Letter*, **2024**. doi: 10.1109/LSENS.2024.3417283. (Q2, IF: 2.2)
7. D Roy, A Ali, S Pal, and **A Adhikary**, "Impedimetric hydrogel sensor for the identification of hexose using machine learning", *IEEE Sensors Journal*, Vol 23(6), pp. 6272-81, **2023**. doi: 10.1109/JSEN.2023.3240750. (Q1, IF: 4.3)
8. A Tapadar, S Sachan, and **A Adhikary**, "Complete design guidelines for fractional-order colpitts oscillator with non-ideal op-amp", *Circuits, Systems, Signal Process.*, *Springer*, Vol 41(10), pp. 5340-65, **2022**. doi: 10.1007/s00034-022-02045-z. (Q2, IF: 1.8)
9. **A Adhikary** and A. Pal, "A six compartments with time-delay model SHIQRD for the COVID-19 pandemic in India: During lockdown and beyond", *Alexandria Engg. J., Elsevier*, Vol 61(2), pp. 1403-12, **2022**. doi:10.1016/j.aej.2021.06.027. (Q1, IF: 6.2)
10. **A Adhikary**, S. Chaudhury and S. Sen, "Optimal design of a fractional order immittance in the second quadrant with wide CPZ", *AEU-Int. J. Electronics Commun., Elsevier*, Vol 130, p. 153563, **2021**. doi:10.1016/j.aeue.2020.153567. (Q2, IF: 3.0)
11. **A Adhikary**, "Characterization, packaging and application of a wide CP zone CNT based fractor", *AEU-Int. J. Electronics Commun., Elsevier*, Vol 127, p. 153441, **2020**. doi: 10.1016/j.aeue.2020.153441. (Q2, IF: 3.0)
12. **A Adhikary**, A Shil, and K Biswas, "Realization of foster structure-based ladder fractor with phase band specification", *Circuits, Systems, Signal Process.*, *Springer*, Vol 39, pp. 2272-92, **2020**. doi:10.1007/s00034-019-01269-w. (Q2, IF: 1.8)
13. **A Adhikary**, J Roy, A Ganeshkumar, S. Bannerjee, K Biswas, "An impedimetric Cu-polymer sensor-based conductivity meter for precision agriculture and aquaculture applications", *IEEE Sensors Journal*, Vol 23(6), pp. 6272-81, **2019**. doi: 10.1109/JSEN.2023.3240750. (Q1, IF: 4.3)

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Administrative Responsibilities —

Current Responsibilities

- Associate HoD of EE Dept,
- Member of Senate, IIT Bhilai
- Member, Institute Purchase Committee, Software

Past Responsibilities

- Associate HoD, MT Dept.
- Convener Departmental PG Committee,
- Convener Departmental UG Committee
- Academic Coordinator MT Dept.
- Faculty in Charge, EE Labs
- Faculty in Charge, Annual Reports
- Faculty in Charge, Website
- Member, Institute Curriculum Revision Committee
- Member, Departmental Purchase Committee
- Member, Committee for Central Instrumentation Facility

Courses Taught

1. Advanced Sensing Techniques
2. Sensor, Measurement, and Instrumentation
3. Control System
4. Advanced Control Theory
5. Optimal Control
6. Circuit and Systems
7. Design of Analog and Mixed Signal Circuits
8. Device and Circuits Lab
9. Sensor Lab
10. Control Lab

Journal Publications (During PhD)

14. **A Adhikary**, S Choudhary, and S Sen, "Optimal design for realizing a grounded fractional order inductor using GIC", *IEEE Trans. Circuits Syst. I*, Vol 65(8), pp. 2411-21, **2018**. DOI: 10.1109/TCSI.2017.2787464. (Q1, IF: 5.1)
15. **A Adhikary**, S Sen, and K Biswas, "Design and Hardware Realization of a Tunable Fractional-Order Series Resonator with High Quality Factor", *Circuits, Systems, Signal Process.*, Springer, Vol 36, pp. 3457-76, **2017**. doi: 10.1007/s00034-016-0469-2. (Q2, IF: 1.8)
16. **A Adhikary**, M Khanra, J Pal, and K Biswas, "Realization of fractional order elements", *Trans. INAE*, Vol 2(2), pp. 41-47, **2017**. doi: 10.1007/s41403-017-0020-1.
17. **A Adhikary**, S Sen, and K Biswas, "Practical realization of tunable fractional order parallel resonator and fractional order filters", *IEEE Trans. Circuits Syst. I*, Vol 63(8), pp. 1142-51, **2016**. doi: 10.1109/TCSI.2016.2568262. (Q1, IF: 5.1)
18. **A Adhikary**, P. Sen, S Sen, and K Biswas, "Design and performance study of dynamic fractors in any of the four quadrants", *Circuits, Systems, Signal Process.*, Springer, Vol 35(6), pp. 1909-32, **2015**. doi:10.1007/s00034-015-0213-3. (Q2, IF: 1.8)

List of International Conferences

1. **A. Adhikary**, D. Karmakar, S. Gupta, and P. Das, "Detection of Early Blight and Septoria Leaf Spot from Tomato Leaf Image using Customized CNN Model with Atrous Convolution Layer", in *IEEE NEWCAS 2025*, Paris, France, **2025**.
2. A. Sahare, A. Tapadar, and **A. Adhikary**, "Robust Fractional Order Notch Filter for Estimating Accurate RS features from ECG signals with Muscle Contraction Noise", in *IEEE NEWCAS 2025*, Paris, France, **2025**.
3. V. Sahu, P. Das, **A. Adhikary**, K. Bandyopadhyay, "Parametric Evaluation and Predictive Modelling of Formability in μ -SPIF Process", in *44th Conf. Int. Deep Drawing Research Group, IDDRG 2025*, Lisbon, Portugal, **2025**.
4. A. Sahare, D. Roy, and **A. Adhikary**, "An Atrous Convolution Based DL Model for Classifications of Aberrant Brain Cells Using MRI", in *2025 IEEE Applied Sensing Conf. (APSCON)*, Hyderabad, **2025**
5. A Awasthi, A Rangana, R Kaushik, D Roy, A Bose, and **A Adhikary**, "Live Demonstration: a Smart Mobile App 'BRIx' for Banana Ripe Stage Detection Using a Decision Tree Model on Image Color Bucket", in *2024 IEEE SENSORS*, Kobe, Japan. **2024**.
6. A Tapadar, A Bose, S Khairnar, Y Mandawat, and **A Adhikary**, "Detection of Urea in Water using Disposable Paper Sensor with FO Colpitts Oscillator", in *2024 IEEE Applied Sensing Conf. (APSCON)*, Goa, India. **2024**. DOI: 10.1109/APSCON60364.2024.10465710.
7. D Roy, A Rangare, R Kaushik, and **A Adhikary**, "Ripening Stage Assessment of Royal Gala Apple using KNN on PCA reduced Bio-impedance Data", in *2024 IEEE Applied Sensing Conf. (APSCON)*, Goa, India. **2024**. DOI: 10.1109/APSCON60364.2024.10466212.
8. A Tapadar and **A Adhikary**, "Design of Fractional Order Hartley Oscillator with Practical Op-amp", in *2023 IEEE 20th India Council Int. Conf. (INDICON)*, Hyderabad, India. **2023**. DOI: 10.1109/INDICON59947.2023.10440960.
9. A Tapadar and **A Adhikary**, "High Sensitivity Milk Adulteration Detector Using Fractional Order Colpitts Oscillator", in *2023 IEEE Conf. AgriFood Electronics (CAFE)*, Torino, India. **2023**. DOI: 10.1109/CAFE58535.2023.10292088.
10. U Bharadwaj, A Tapadar, and **A Adhikary**, "Improvement in transient and steady state response of a buck converter by designing optimal fractional order controllers using GA and MA", in *2023 14th Int. Conf. Computing Communication Networking Technol. (ICCCNT)*, Delhi, India. **2023**. DOI: 10.1109/ICCCNT56998.2023.10307677.
11. A Tapadar and **A Adhikary**, "Inductor-less FO Low-pass Notch Filter with High Q and High Attenuation for ECG Signal Processing", in *2023 IEEE Int. Instrumentation Measurement Technol. Conf. (I2MTC)*, Kuala Lumpur, Malaysia. **2023**. DOI: 10.1109/I2MTC53148.2023.10176035.

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Outreach Activities

- 14 Invited Lectures on topics of Sensor and Control System
- Invited Reviewers of 'Virtual Lab', by IIT Delhi, in 2021, 22, 24.
- Reviewer of IEEE TIM, IEEE Sensors J and IEEE flagship conferences.
- Organized 6 FDP and 8 Short Term Courses under TEQIP-III.

Graduated PhD, MTech and BTech Hons. Students

- Agniv Tapadar **PhD**, 2024
Thesis: Fractional Order Colpitts Oscillators for Impedimetric Analysis of Bio and Chemical Sensors
- Sathyam Sharma **MTech**, 2024
Thesis Title: Development of machine-learning nondestructive ripeness detection system for banana using computer vision
- Ujjwal Upadhyay **MTech**, 2023
Thesis: Design of Optimal FO Controllers for Improvement in Transient and Steady State Response of a Buck Converter
- Aakashdeep Kurre **MTech**, 2021
Thesis: Study of Power Converters in Fractional Order Domain
- Krishnan Dwivedi **MTech**, 2021
Thesis Topic: Design and Realization of OTA based FO Filters
- Swaninda Ghosh **BTech Hons**, 2021
Thesis: Consensus controlled framework for Multi-agent System and Formation Control in a Multi-agent System
- Rohit Bose **BTech Hons**, 2021
Thesis: Consensus Algo for multi-agent system and Flocking Theory
- Harshit Bansal **BTech Hons**, 2020
Thesis: Optimal Design of a FO Immittance in the Third Quad.
- Shristi Sachan **BTech Hons**, 2019
Thesis: Fractional Order Colpitts and Hartley Oscillator

Conference Publications (Contd.)

12. D Roy and **A Adhikary**, "Detection of Ripening stage of Banganapalle Mango using KNN method on PCA-reduced EIS Data", in *2023 IEEE Int. Instrumentation Measurement Technol. Conf. (I2MTC)*, Kuala Lumpur, Malaysia. **2023**. DOI: 10.1109/I2MTC53148.2023.10175886.
13. A Tapadar and **A Adhikary**, "Optimal position of fractional order inductor in fractional order hartley oscillator", in *2022 Int. Conf. Futuristic Technol. (INCOFT)*, Belgaum, India. **2022**. DOI: 10.1109/INCOFT55651.2022.10094388.
14. **A Adhikary** and D Roy, "Fractional order parameter estimation from the nyquist plots with two distinct lobes", in *2020 First Int. Conf. Power, Control Computing Technol. (ICPC2T)*, Raipur, India. **2020**. DOI: 10.1109/ICPC2T48082.2020.9071455.
15. **A Adhikary**, J Roy and K Biswas, "Performance study of a two-electrode type aqueous conductivity sensor for smart farming", in *2019 IEEE Int. Instrumentation Measurement Technol. Conf. (I2MTC)*, Auckland, New Zealand. **2019**. DOI: 10.1109/I2MTC.2019.8826887.
16. **A Adhikary**, S Sen and K Biswas, "Realization and study of a fractional order resonator using an obtuse angle fractor", in *2016 IEEE Students' Technol. Symp. (TechSym)*, Kharagpur, India. **2016**. DOI: 10.1109/TechSym.2016.7872667.
17. **A Adhikary**, A Ganeshkumar, S. Bannerjee, S. Sen, and K Biswas, "Modelling and performance improvement of phase-angle-based conductivity sensor", in *2016 IEEE First Int. Conf. Control, Measure. Instrum. (CMI)*, Kolkata, India. **2016**. DOI: 10.1109/CMI.2016.7413779.
18. **A Adhikary**, M. Khanra, S. Sen, and K Biswas, "Realization of a carbon nanotube based electrochemical fractor", in *2015 IEEE int. symp. circuits systems (ISCAS)*, Lisbon, Portugal. **2015**. DOI: 10.1109/ISCAS.2015.7169150.
19. **A Adhikary**, U Kumar, and K Biswas, "A Phase Angle Measurement Based Conductivity Sensor for Low Conductance Solution", in *2015 Symp. Advances Control & Instrum. (SACI 2014)*, Mumbai, India. **2014**.
20. **A Adhikary**, M Khanra, and K Biswas, "Design of FPGA based digital controller for 2nd and higher order systems", in *2012 Student Conf. Engg. Syst. (SCES 2012)*, Allahabad, India. **2012**. DOI: 10.1109/SCES.2012.6199059.

Book Chapters

1. A Tapadar, F A Khandey, S Sen, and **A Adhikary**, "Fractional calculus in electronic circuits: A review", in *Fractional Order Systems*, Academic P., pp. 441-482, **2022**.
2. **A Adhikary**, "CNT-based fractors in all four quadrants: design, simulation, & practical applications", in *Fractional-Order Design*, Academic P., pp. 235-274, **2022**.
3. **A Adhikary** and K Biswas, "Four-quadrant fractors and their applications in fractional order circuits", in *Applications in Engineering, Life and Social Sciences, Part B*, Walter de Gruyter GmbH Co KG, pp. 33-62, **2019**.

List of Current Collaborators

1. Dr Suchetan Pal, Dept Chem, IIT Bhilai
2. Dr Soumajit P, Dept CSE, IIT Bhilai
3. Dr Kaushik B, Dept ME, IIT Bhilai
4. Dr A Bose, Dept EE, NIT Raipur
5. Dr Saikat Sahoo, MT, IIT Bhilai
6. Dr Siddharth K, MT, IIT Bhilai

Dr Avishek Adhikary
Durg, May 2025